

# Data Structures Overview

## Arrays

An array is a collection of elements stored at contiguous memory locations. Access is  $O(1)$  by index.

## Linked Lists

A linked list is a linear data structure where elements point to the next node. Insertion is  $O(1)$  at the head.

## Stacks

A stack follows LIFO order. Push and pop operations both run in  $O(1)$  time complexity.

## Queues

A queue follows FIFO order. Enqueue and dequeue both run in  $O(1)$  time complexity.